Design and Modeling of Complex Software Architectures

Lab Assignment 03, DDD: Bounded Contexts & Context Mapping

Application Scenario 4: Instant Mobility Platform

Team: The Winx | Summer Semester 2026

1 Overview

Domain-Driven Design (DDD) Strategic Design decomposes a complex system into Bounded Contexts, explicit boundaries within which a specific domain model applies, and a consistent Ubiquitous Language is used. This document identifies the five bounded contexts of the Instant Mobility Platform and maps the relationships between them using Vernon's context mapping notation.

Each bounded context corresponds to one microservice in the system architecture. Domain concepts that appear in multiple contexts are represented with different attributes in each context; only the data relevant to that context of responsibility is present.

2 Identified Bounded Contexts

Five bounded contexts were identified by grouping domain concepts around cohesive business capabilities with clear ownership boundaries and distinct ubiquitous languages.

ID	Context	Responsibility	Key Concepts
BC-01	**Identity & Access**	Authentication & account management for Users and Providers.	User (identity), Provider (identity)
BC-02	**Fleet Management**	Vehicle CRUD, pricing, restrictions, location, and status tracking for vehicles.	Provider (owner), Vehicle, VehicleType, VehicleStatus, BillingModel
BC-03	**Booking**	Vehicle search, booking lifecycle, cost computation, and payment trigger.	User (customer), Vehicle (availability), Booking, BookingStatus
BC-04	**Payment**	Financial transaction processing for completed bookings.	Booking (ref), Payment, PaymentStatus
BC-05	**Rating**	User feedback on vehicles and providers after completing bookings.	User (author), Vehicle (ref), Provider (ref), Booking (ref), Rating

2.1 Context Details

BC-01 Identity & Access (Purple)

Responsibility: Handles registration, authentication and credential management for both Users and Providers. Acts as the single source of truth for identity across the entire platform.

Domain Concept	Role in this context
User	Identity view, credentials, email, name, registeredAt. No booking or rating data lives here.
Provider	Identity view, credentials, companyName, contactName. No fleet data lives here.

Shared concepts: User and Provider appear in other contexts as well, but only their identity attributes (ID, email, name) are shared, not the full entity.

Ubiquitous Language:

- Registration, the act of creating a new account on the platform.
- Authentication, verifying that the supplied credentials match a stored account.
- Session, a time-limited proof that a principal has been authenticated.
- Principal, a generic term for any authenticated party (User or Provider).

BC-02 Fleet Management (Teal)

Responsibility: Manages the full lifecycle of vehicles offered by Providers: creation, updates, deletion, pricing configuration, restriction rules, and real-time availability and GPS location tracking.

Domain Concept	Role in this context
Provider	Fleet-owner view, which vehicles belong to this provider, fleet status overview.
Vehicle	Full entity, type, status, location, pricing, all restriction attributes.
VehicleType	Enum: E_SCOOTER | BICYCLE | E_BIKE | E_CAR.
VehicleStatus	Enum: AVAILABLE | BOOKED, updated when a vehicle is booked or released through the Booking context.
BillingModel	Enum: PER_HOUR | PER_KILOMETER, set per vehicle by the Provider.

Shared concepts: Vehicle is shared into the Booking context (availability/location view) and into the Rating context (reference for what was rated).

Ubiquitous Language:

- Fleet, the collection of all vehicles owned by a single Provider.
- Availability, whether a vehicle can currently be booked.
- Restriction, a rule that constrains who may book a vehicle and for how long.
- Pricing, the combination of pricePerUnit and billingModel for a vehicle.

BC-03 Booking (Blue)

Responsibility: Owns the core rental lifecycle: searching for available vehicles by location, filtering by type and restrictions, creating a booking, tracking the active ride, computing the final cost on completion, and triggering payment.

Domain Concept	Role in this context
User	Customer view, who is making the booking. User ID and relevant profile data, such as age restrictions, are used.
Vehicle	Availability view, current location, status, pricePerUnit, billingModel, restrictions. Read from Fleet Management.
Booking	Full entity, start/end times, GPS coordinates, distanceKm, totalCost, status.
BookingStatus	Enum: ACTIVE | COMPLETED | CANCELLED.

Shared concepts: Booking is shared (read-only reference) into the Payment context (to trigger and settle payment) and into the Rating context (to check that a booking is COMPLETED before a rating is allowed).

Ubiquitous Language:

- Search, finding vehicles within a given radius of the user's position.
- Booking, a confirmed reservation of a vehicle for immediate use.
- Active ride, the period between booking start and booking end.
- Cost computation, deriving totalCost from pricePerUnit, billingModel, duration, and distance.

BC-04 Payment (Amber)

Responsibility: Processes and records financial transactions for completed bookings. Receives a payment trigger from the Booking context, charges the user, and records the payment status (PENDING, PAID, or FAILED). Owns no booking logic.

Domain Concept	Role in this context
Booking	Reference only, bookingId and totalCost are used to create the Payment record.
Payment	Full entity, amount, paymentMethod, paidAt, status.
PaymentStatus	Enum: PENDING | PAID | FAILED.

Shared concepts: No domain concepts from Payment are shared outward. Payment is purely a downstream consumer of Booking.

Ubiquitous Language:

- Payment, a financial transaction tied to one completed booking.
- Charge the act of debiting the user's chosen payment method.
- Settlement confirmation that the payment has been successfully processed.

BC-05 Rating (Coral / Red)

Responsibility: Collects and stores user feedback (scores and comments) for a Vehicle and its Provider after a booking reach COMPLETED status. Enforces the rule that only one rating is allowed per booking.

Domain Concept	Role in this context
User	Author view, who wrote the rating. Only userId is stored.
Vehicle	Rated subject, vehicleId used to identify the rated vehicle. Read from Fleet Management.
Provider	Rated subject, providerId resolved from the rated vehicle through Fleet Management.
Booking	Reference, bookingId used to validate that the booking is COMPLETED and that no rating exists yet.
Rating	Full entity, vehicleScore, providerScore, comment, createdAt.

Shared concepts: No domain concepts from Rating are shared outward.

Ubiquitous Language:

- Rating, a user's evaluation of a vehicle and provider after a completed ride.
- Score, a numeric value (1–5) representing quality.
- Review the optional free text comment accompanying the scores.

3 Context Map (Vernon's Notation)

The context map below documents all integration relationships between the five bounded contexts. For each relationship the upstream (U) and downstream (D) context are identified, and one or more context mapping patterns from Vaughn Vernon's DDD are applied.

3.1 Pattern Reference

Abbreviation	Pattern	Meaning
`OHS`	Open Host Service	The upstream context publishes a well-defined, stable service protocol (API) that any downstream can consume. Avoid tight coupling to internal implementation details.
`PL`	Published Language	A well-documented shared exchange format (e.g. JSON schema, protobuf) used by the OHS. Allows downstream consumers to integrate without knowing the upstream's internal model.

`ACL`	Anti-Corruption Layer	A translation layer in the downstream context that converts the upstream model into the downstream's own domain model. Protects the downstream from upstream changes and keeps the domain clean.
`C/S`	Customer / Supplier	A negotiated relationship where the downstream (customer) can influence the upstream (supplier) interface through planning discussions. The supplier gives priority to the customer's needs.
`CF`	Conformist	The downstream adopts the upstream model without translation. Chosen when the upstream model is stable and well-designed enough that translation would add cost without benefit.

3.2 Relationships

Upstream (U)	Downstream (D)	U-Pattern	D-Pattern	Rationale
Identity & Access	Booking	**OHS / PL**	**ACL**	Booking must verify User identity and retrieve age for restriction checks. Identity & Access provides the same OHS/PL. Booking uses an ACL to translate the IAC User identity into its own lightweight customer representation.
Fleet Management	Booking	**OHS / PL**	**ACL**	Booking queries Fleet Management for available vehicles (by location, type, and restrictions) and reads vehicle pricing data to compute ride costs. Fleet Management publishes a stable OHS with a clearly defined API. Booking uses an ACL so that internal changes to Fleet Management's vehicle model do not propagate into Booking's domain model.
Booking	Payment	**Customer / Supplier (C/S)**	**Conformist (CF)**	After a Booking is COMPLETED, the Booking context triggers payment by passing the bookingId and totalCost to the Payment context. This is a negotiated Customer/Supplier relationship, Booking (supplier) and Payment (customer) collaborate on the interface. Payment is Conformist: it accepts and uses Booking's data structure (bookingId, totalCost) without translation, simplifying integration.
Booking	Rating	**Customer / Supplier (C/S)**	**Conformist (CF)**	Rating is Conformist because it uses Booking's bookingId and BookingStatus information directly instead of maintaining its own booking lifecycle model.
Fleet Management	Rating	**OHS / PL**	**ACL**	Rating needs a stable reference from Vehicle to Provider so that the rating can store vehicleId and providerId. Fleet

				Management exposes this vehicle-provider information via its OHS. Rating uses an ACL to translate the Fleet data into its own rating model, decoupling Rating from future changes to Fleet Management's vehicle model.

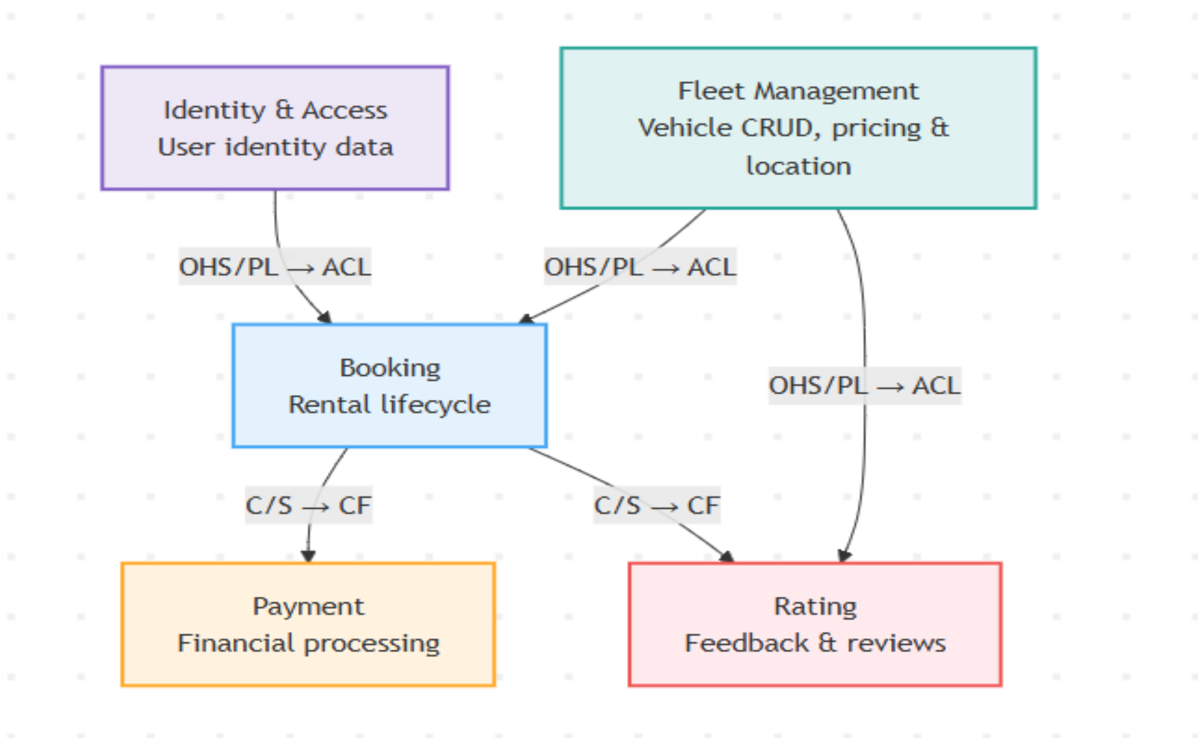
3.3 Context Map Narrative

Identity & Access acts as an upstream identity service. Booking retrieves user identity data from Identity & Access using the OHS/PL it provides, wrapping the result in an ACL so that identity changes never pollute the Booking domain model. This prevents Booking from re-implementing identity logic.

Fleet Management sits in an upstream position relative to Booking and Rating. It publishes vehicle and provider data via an OHS. Booking uses this to locate available vehicles and to retrieve pricing for cost computation. Rating uses it to resolve the providerId for the rated vehicle. Both protect themselves via an ACL.

Booking is the central operational context; it orchestrates the ride's lifecycle. It is simultaneously a downstream consumer of IAC and Fleet Management, and an upstream supplier to Payment and Rating. The relationship with Payment and Rating is Customer/Supplier. Both downstream contexts had input into the interface during design. Payment and Rating are Conformist because Booking's output data, such as bookingId, totalCost, and BookingStatus, are stable and well-defined.

Rating is a leaf context. Payment is mainly downstream of Booking, but its payment status can be read back for booking/payment display.



Context mapping decisions:

OHS/PL → ACL used where the upstream is a stable platform service (Identity & Access, Fleet Management). The ACL on the downstream side means internal upstream changes never leak into other contexts.

Customer/Supplier → Conformist used between Booking→Payment and Booking→Rating because both downstream contexts had a say in the interface design (C/S), and Booking's output data is clean enough that no translation layer is needed (CF).